

PAPER_292: Crab Pulsar 60-Second UQFF Resonance Window — $f_{\text{osc}} = 1812$ Hz Spin-to-Vacuum DPM Lock

Authors: Daniel T. Murphy

Series: UQFF Whitepaper Series — Session 82

Module: CRAB_RESONANCE_UQFF_MODULE.cpp (24th C++ module)

Date: March 2026

Abstract

This paper derives the first UQFF pulsar spin-to-vacuum coupling mechanism, based on the observation that the Crab pulsar spin frequency of 30.2 Hz, integrated over a standard 60-second timing window, produces a resonance frequency $f_{\text{osc}} = 1812$ Hz that locks fractionally to the DPM vacuum hierarchy. The resulting angular frequency $\omega_{\text{pulsar}} = 11,385$ rad/s defines a new oscillatory modulation term in the UQFF Crab module, added to the inherited cosmic-age standing/ traveling oscillatory structure of PAPER_288. The DPM lock ratio $\text{pulse_lock} = f_{\text{osc}}/f_{\text{DPM}} = 1.812 \times 10^{-9}$ and the DPM lock amplitude $A_{\text{pulsar}} = 1.812 \times 10^{-19}$ m are computed.

1. Background: Crab Pulsar Timing

The Crab Pulsar (PSR J0534+2200) is one of the most precisely timed astrophysical objects:

Parameter	Value	Notes
Spin frequency	$f_{\text{pulsar}} = 30.2$ Hz	Period $P = 33.1$ ms
Period derivative	$\dot{P} = 4.2 \times 10^{-13}$ s/s	Spin-down rate
Characteristic age	$\tau = P/(2\dot{P}) = 1250$ yr	Spin-down age
Spin-down luminosity	$\dot{E} = 4\pi^2 \dot{P} P^3 \approx 5 \times 10^{31}$ W	Powers the PWN
Standard timing window	60 seconds	Dispersion-measure epoch

2. The 60-Second UQFF Resonance Window

2.1 Physical Motivation

In pulsar timing, a 60-second integration window is the standard epoch for folding pulsar emission to construct a mean profile and measure time-of-arrival (TOA). During 60 seconds, the Crab pulsar emits exactly:

$$N_{\text{pulses}} = f_{\text{pulsar}} \times 60 \text{ s} = 30.2 \times 60 = 1812 \text{ pulses}$$

In UQFF theory, these 1812 discrete pulses per 60-second window create a coherent resonance frequency — the rate at which the pulsar’s electromagnetic and particle output couples to the plasmonic vacuum on a 1-minute integration timescale.

2.2 UQFF Resonance Frequency

$$f_{\text{osc}} = f_{\text{pulsar}} \times 60 = 30.2 \times 60 = 1812 \text{ Hz}$$

$$\omega_{\text{pulsar}} = 2\pi f_{\text{osc}} = 2\pi \times 1812 = 11,385.0 \text{ rad/s}$$

3. DPM Vacuum Lock Ratio

The fractional coupling between the 60s resonance window frequency and the DPM mode:

$$\text{pulse_lock} = \frac{f_{\text{osc}}}{f_{\text{DPM}}} = \frac{1812}{10^{12}} = 1.812 \times 10^{-9}$$

This dimensionless ratio represents the fractional vacuum-lock — the DPM mode oscillates $10^{12}/1812 = 5.517 \times 10^8$ times faster than the pulsar resonance window. In spectral terms:

$$\log_2 \left(\frac{f_{\text{DPM}}}{f_{\text{osc}}} \right) = \log_2(5.517 \times 10^8) = 29.0 \text{ octaves}$$

The DPM-to-pulsar frequency ladder spans exactly 29 octave steps.

4. Synchrotron Scale Comparison

The synchrotron emission angular frequency in the Crab is encoded in $\omega_{\text{osc}} = 10^{15}$ rad/s. The ratio to the pulsar resonance angular frequency:

$$\frac{\omega_{\text{osc}}}{\omega_{\text{pulsar}}} = \frac{10^{15}}{11,385} = 8.785 \times 10^{10}$$

The synchrotron scale is **88 billion times** higher in angular frequency than the 60s pulsar window. This large ratio defines the dynamic range of the Crab's UQFF oscillatory spectrum.

5. DPM Lock Amplitude

The coupling amplitude of the pulsar modulation to the UQFF oscillatory term is set by the pulse_lock ratio times the base oscillation amplitude:

$$A_{\text{pulsar}} = \frac{f_{\text{osc}}}{f_{\text{DPM}}} \times A_{\text{amp}} = 1.812 \times 10^{-9} \times 10^{-10} \text{ m} = 1.812 \times 10^{-19} \text{ m}$$

This is a sub-nuclear length scale (nuclear radius $\sim 10^{-15}$ m $\rightarrow A_{\text{pulsar}} = 10,000\times$ smaller than nuclear). It represents the fractional UQFF vacuum displacement modulated by the pulsar timing window.

6. Modified Oscillatory Term (PAPER_288 + PAPER_292)

The standard cosmic-age standing/traveling wave oscillatory term (PAPER_288) is augmented by the pulsar modulation:

$$a_{\text{osc}}(t) = \underbrace{2A \cos(kx) \cos(\omega_{\text{osc}} t)}_{\text{standing (PAPER_288)}} + \underbrace{\frac{2\pi}{13.8} A \text{Re}[e^{i(kx - \omega_{\text{osc}} t)}]}_{\text{cosmic-age traveling (PAPER_288)}} + \underbrace{A_{\text{pulsar}} \cos(\omega_{\text{pulsar}} t)}_{\text{pulsar DPM lock (PAPER_292)}}$$

With $A = 10^{-10}$ m, $\omega_{\text{osc}} = 10^{15}$ rad/s, $T_{\text{cosmic}} = 13.8$ Gyr, $\omega_{\text{pulsar}} = 11,385$ rad/s:

Component	Amplitude	Angular Frequency
Standing (PAPER_288)	$2A = 2 \times 10^{-10}$ m	$\omega_{\text{osc}} = 10^{15}$ rad/s
Traveling (PAPER_288)	$(2\pi/13.8) \times A = 4.55 \times 10^{-11}$ m	$\omega_{\text{osc}} = 10^{15}$ rad/s
Pulsar lock (PAPER_292)	$A_{\text{pulsar}} = 1.812 \times 10^{-19}$ m	$\omega_{\text{pulsar}} = 11,385$ rad/s

The pulsar contribution amplitude is $\sim 10^9 \times$ smaller than the standing wave — but it operates at **nine orders of magnitude lower angular frequency**, making it the dominant **long-timescale** modulation of the Crab UQFF oscillatory structure.

7. The “60× Octave” and Pulsar Rotation

The factor 60 is not arbitrary in UQFF:

- $60 = 2^2 \times 3 \times 5$ — the smallest integer encompassing all three fundamental resonance numbers (2: binary systems/standing-wave, 3: 3-body gravity, 5: five-mode oscillatory cascade)
- $\log_2(60) = 5.907$ octaves — f_{osc} is ~ 5.9 octaves above f_{pulsar} , encoding the transition from pulsar-spin to timing-epoch scale
- $1812 \text{ Hz} = f_{\text{osc}}$: interestingly, $1812 \times 1000 = 1.812 \text{ GHz} \approx 21\text{-cm HI line} / 1.42 \text{ GHz} \times 1.27$ (connecting pulsar resonance window to galactic spiral arm HI emission — see PAPER_274)

8. Observational Implications

UQFF PAPER_292 Prediction: The DPM vacuum modulation at $f_{\text{osc}} = 1812$ Hz should produce a submillimeter polarization fluctuation in the Crab’s PWN with period $T_{\text{pulsar}} = 1/f_{\text{osc}} = 0.552$ ms. This is 16.7 times the pulse period ($33.1 \text{ ms} / 0.552 \text{ ms} \times \dots$ wait, corrected: $T_{\text{osc}} = 1/1812 = 0.552$ ms which is shorter than the 33.1 ms pulse period). Future VLBI timing observations at sub-millisecond resolution may detect this modulation as a periodic oscillation in the DM-corrected pulse profile.

9. Comparison with Prior UQFF Pulsar Treatment

Prior UQFF modules referenced the Crab as an external reference object: - **PAPER_256 (Session 72d)**: CrabNebulaM1FUBiCalculator — used Crab as compact $r=10^4$ m, $B_0=10^{-4}$ T $\rightarrow$ DPM geometry-dependency discovery (compact vs diffuse) - **PAPER_292 (This paper)**: First UQFF treatment of the Crab **pulsar spin frequency** as a resonance driver — distinct from the nebular radius-based computation of PAPER_256

10. Wolfram KB Registration

```
CRAB_UQFF:f_osc=30.2*60=1812 Hz; omega_pulsar=2*Pi*f_osc=11385 rad/s;  
A_pulsar=(f_osc/f_DPM)*A_amp=1.812e-19 m; pulse_lock=1.812e-9;  
a_osc+=A_pulsar*Cos[omega_pulsar*t] [PAPER_292 pulsar DPM lock]
```

Session 82 — 24th C++ UQFF Module — PAPER_292 of 1000